NETFLOW

F L O W M E A S U R E M E N T · N T - T U F S E R I E S

WALL MOUNTED ULTRASONIC FLOWMETER

NT-TUF-S Series — Clamp-on, Insertion and Pipe type ultrasonic flow measurement

The NT-TUF-S flowmeter can be applied to a wide range of long-term online measurement applications. Convertor protection class IP67, transducer protection class IP68. Accommodates ultra-pure liquids, potable water, chemicals, raw sewage, reclaimed water, river water, plant effluent, etc.

±1% MEASUREMENT ACCURACY	**IP67 / IP68** CONVERTOR / TRANSDUCER PROTECTION
DN15 — DN6000 PIPE SIZE RANGE	**-30 … +160 °C** FLUID TEMPERATURE RANGE

- Measurement accuracy ±1%
- Convertor protection class IP67, transducer protection class IP68
- Wide measurement range, pipe size from DN15mm to DN6000mm
- Wall mounted, DIN-rail mounted, explosion-proof box mounted
- Connect temperature transducer for heat/energy measurement

INSTALLATION TYPES

Clamp-on

No need to cut off water, no pressure loss. Easy installation. Heat/energy measurement with clamp-on temperature transducer.

Insertion

No need to cut off water, no pressure loss. Stable and reliable for long-term operation. Heat measurement with Pt100.

Pipe

High accuracy and stability. Heat measurement with Pt100. Requires pipe cut-off.

FLOW TRANSDUCERS

TYPE	MODEL	RANGE	TEMP.	SIZE
Clamp-on Small	TS-2	DN25–DN100	-30–90 °C	45×25×32mm
Clamp-on Medium	TM-1	DN50–DN700	-30–90 °C	64×39×44mm
Clamp-on Large	TL-1	DN300–DN6000	-30–90 °C	97×54×53mm
Hi-Temp Small	TS-2-HT	DN25–DN100	-30–160 °C	45×25×32mm
Hi-Temp Medium	TM-1-HT	DN50–DN700	-30–160 °C	64×39×44mm
Hi-Temp Large	TL-1-HT	DN300–DN6000	-30–160 °C	97×54×53mm
Insertion Std	TC-1	DN80–DN6000	-30–160 °C	190×80×55mm
Insertion Long	TC-2	DN80–DN6000	-30–160 °C	335×80×55mm
Pipe Pi	G1	DN15–DN32	-30–160 °C	SUS304 thread
Pipe Std	G2	DN40–DN1000	-30–160 °C	Carbon steel thread

TEMPERATURE TRANSDUCERS

TYPE	MODEL	RANGE	TEMP.	CUT-OFF	ACCURACY
Clamp-on Pt100	CT-1	>DN50	-40–160 °C	No cut-off	±0.8 °C
Insertion Pt100	TCT-1	>DN50	-40–160 °C	Cut-off req.	±0.8 °C
Insertion w/ pressure	PCT-1	>DN50	-40–160 °C	No cut-off	±0.8 °C
Small pipe Pt100	SCT-1		-40–160 °C	Cut-off req.	±0.8 °C

SD CARD

Convenience for long-term and large data storage. Available for tabulation, statistical analysis, printing reports and production flow curve using data analysis software.

Principle	Transit-time ultrasonic flowmeter
Accuracy	±1%
Display	2×20 character LCD with backlight, English/Turkish
Signal Output	1× 4–20 mA (load 0–1 kΩ, accuracy 0.1%)
Pulse Output	1× OCT pulse (width 6–1000 ms, default 200 ms)
Relay Output	1× relay output
Signal Input	3× 4–20 mA input, accuracy 0.1%
Data Interface	Isolated RS485, MODBUS protocol
Pipe Material	Steel, Stainless, Cast iron, Copper, Cement, PVC, Aluminium, FRP
Pipe Diameter	DN15–DN6000 mm
Straight Pipe	Upstream 10D, downstream 5D, 30D from pump
Liquid Type	Single liquid that can transmit sound wave
Temperature	-30 °C – +160 °C
Turbidity	< 10000 ppm, minimal bubbles
Flow Rate	0 – ±7 m/s
Working Temp	Convertor: -20–60 °C · Transducer: -30–160 °C
Humidity	Convertor: 85% RH
Power Supply	DC 8–36 V or AC 85–264 V (optional)
Power Consumption	1.5 W
Dimensions	132×150×85 mm (convertor)

ORDERING CODE

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NT-TUF-S ---- [Transducer] - DN[Diameter]mm - [Pipe Material] - [Pressure]MPa - [Cable]m - [Temp. Transducer] - [SD]
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Standard configuration: NT-TUF-S (convertor) + TM-1 + 5M cable + Steel belt (DN200) + Couplant (1 pc)